

## Trainings ▾

### General Information

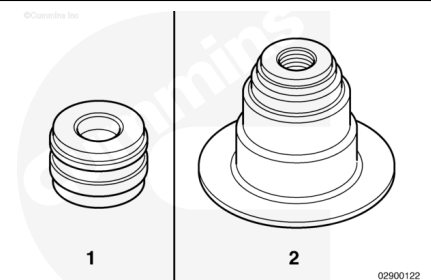
The following procedure is for removing the valve stems seals with the cylinder head installed. For removing the valve stem seals with the cylinder head removed, Refer to Procedure 002-004 in Section 2 (</qs3/pubsys2/xml/en/procedures/40/40-002-004-tr.html>).

**Note :** This procedure can also be used for removing valve springs, valve spring retainers and valve collets with the cylinder head installed.

For the engines covered by this procedure, different types and color valve stem seals are used. When replacing the valve stem seals, always replace like parts.

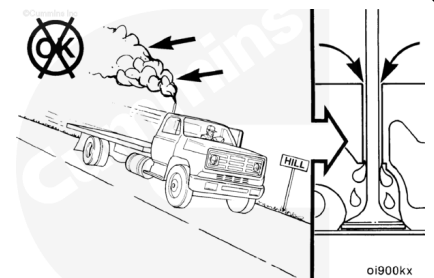
There are two types of valve stem seals used:

1. "Drive-On" Seal
2. "Top-Hat" Seal

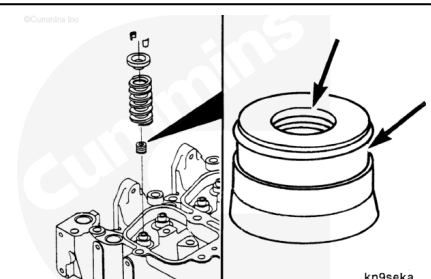


#### Valve Seal - Wear

Worn valve seals are typically detected by excessive smoke at idle or when the engine is unloaded when the vehicle is going downhill. Verify the condition by removing the valve spring and inspecting the valve seals.



Hardening of the material and wear or damage to the sealing surfaces will cause the valve seal to leak.



## Preparatory Steps

### **⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**



- Disconnect the batteries. See equipment manufacturer service information.
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3  
(/qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html).

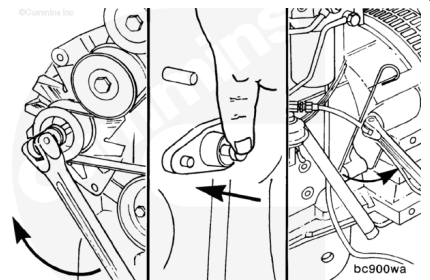
## Remove

**Note :** In order to remove the valve springs with the cylinder head installed, the piston of the cylinder being worked on must be brought to TDC to support the valves.

Use the barring tool, Part Number 3824591. Locate top dead center (TDC) for cylinder number 1 by barring the crankshaft slowly while pressing on the engine timing pin.

Service Tip: If the barring tool is inaccessible, in some applications the alternator pulley can be used to bar the engine over.

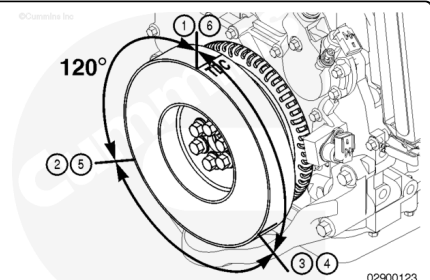
When the timing pin engages in the hole in the camshaft gear, cylinder number 1 is at TDC on the compression stroke.



With the timing pin locked at TDC for the number 1 cylinder, mark the damper for the location of TDC for the other cylinders.

For six cylinder engines:

- Mark the vibration damper damper every 120 degrees with a marker directly on the damper or to a piece of masking tape applied directly around the damper.



Service Tip: A protractor, camshaft degree wheel or angle/level indicator, part number 3375855, can be used to locate 120-degree increments around the vibration damper.

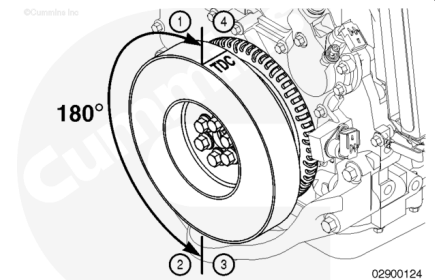
- Mark the damper with the TDC indicator for each cylinder as shown. Two cylinders correspond to each 120-degree line.

For four cylinder engines:

- Mark the vibration damper every 180 degrees with a marker directly on the damper or to a piece of masking tape applied directly around the damper.

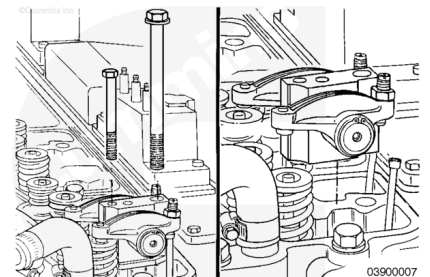
Service Tip: A protractor, camshaft degree wheel or angle/level indicator, part number 3375855, can be used to locate 180-degree increments around the vibration damper/crankshaft pulley.

- Mark the damper with the TDC indicator for each cylinder as shown. Two cylinder correspond to each 180-degree line.



For engines with 4 valves per cylinder, remove the rocker lever assemblies and crossheads. Refer to Procedure 003-008 (</qs3/pubsys2/xml/en/procedures/40/40-003-008-tr.html>) in Section 3 and Refer to Procedure 002-001 in Section 2 (</qs3/pubsys2/xml/en/procedures/40/40-002-001-tr.html>).

For engines with 2 valves per cylinder, remove the rocker lever assemblies. Refer to Procedure 003-008 in Section 3 (</qs3/pubsys2/xml/en/procedures/40/40-003-008-tr.html>).



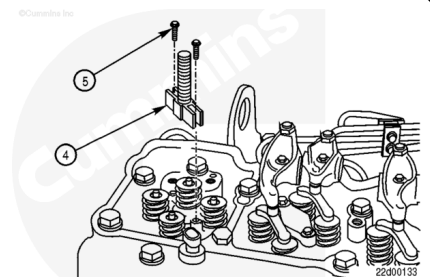
4 Valves per Cylinder:

Compress the valve springs. Use the valve spring compressor, Part Number 3164329.

Position the replacer screw (4) above the injector bore and install the two capscrews (5) in the cylinder head where the hold-down clamp screws were removed.

Tighten the capscrews (5).

**Torque Value:** 5 n•m [ 44 in-lb ]



**Note :** The valves are not evenly spaced from the injector bore. It is important to align the slots in the

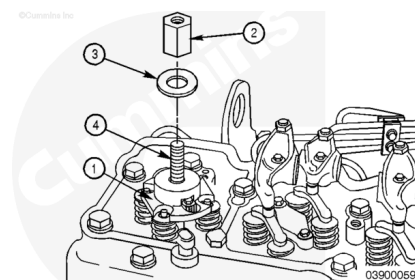
**valve spring compressor plate with the valve springs.**

Apply anti-seize lubricant to the replacer screw (4) threads.

**Always** read and follow label precautions.

Position the valve spring compressor plate (1) on the replacer screw (4) and align the slots in the valve spring compressor plate with the valve springs.

Install the washer (3) and nut (2) on the replacer screw (4).



**⚠ WARNING ⚠**

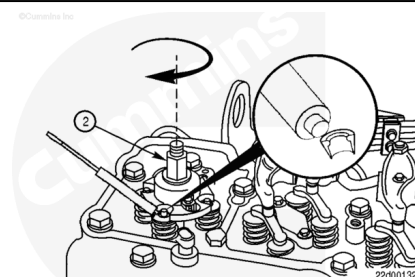
**Valve springs are under tension and can act as projectiles if released. To reduce the possibility of eye injury, wear safety glasses with side shields.**

Turn the nut (2) **clockwise** to compress the valve springs.

Continue turning the nut (2) **clockwise** until the valve collets can be removed. Use a magnetic tool, such as the end of a magnetic screwdriver.

**Note :** Because there is a gap between the top of the piston and the valve face, it may be necessary to use a second magnet to hold the valve stem up to remove the valve collets.

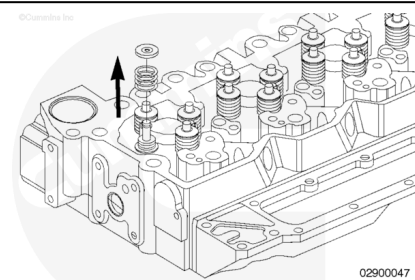
Remove the valve collets and the valve spring compressor service tool.



**⚠ CAUTION ⚠**

**With the valve collets, valve springs, and valve spring retainers removed, do not rotate the engine. Rotating the engine will allow the valves to drop into the cylinder requiring the cylinder head to be removed or possible engine damage.**

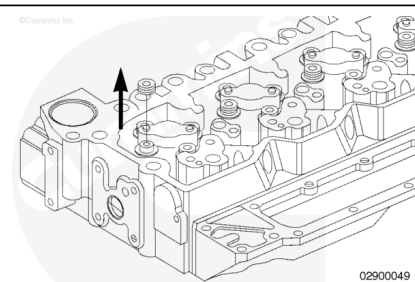
Remove the four valve spring retainers and valve springs.



Use boot pliers, Part Number 3163293, to remove the valve stem seals.

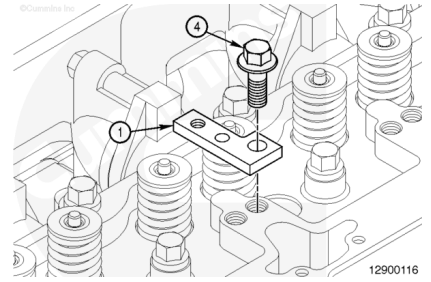
**Note :** The valve stem seals can be installed by hand. The installation tool will aid with installing the valve stem seals, but is not mandatory.

Note the type, and location of the seal, then discard the old seal.



**⚠ CAUTION ⚠**

**Do not use an impact tool for any part of this procedure.  
Doing so can damage the tool and/or the engine.**

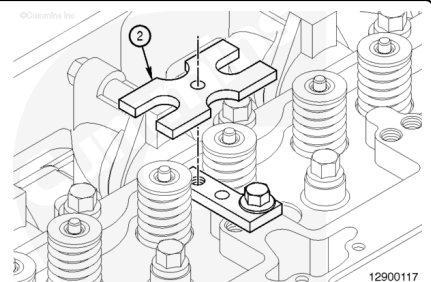


2 Valves Per Cylinder:

Compress the valve springs with valve spring compressor tool kit, Part Number 3163292.

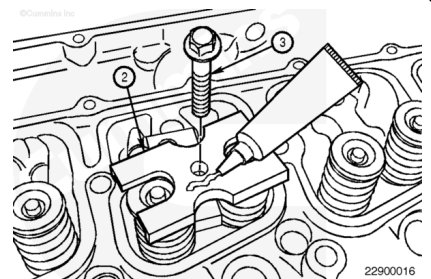
Position the lower support plate (1) with the part number up, and install the 10-mm capscrew (4) through the plate and secure to the cylinder head. Tighten the capscrew finger-tight.

Position the spring compression plate (2) onto the valve springs with the part number up.



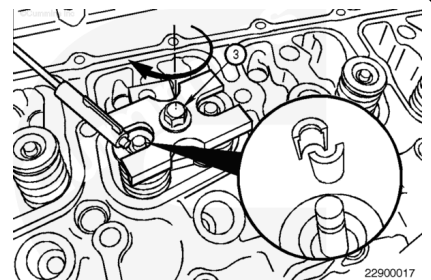
Apply grease to the top of the spring compression plate (2) and underneath of the capscrew (3) head and install the 10-mm (3) capscrew.

Tighten the capscrew finger-tight (3).



**⚠ WARNING ⚠**

**Wear appropriate eye and face protection when performing this procedure. Springs are under tension and can act as a projectile if released, causing personal injury.**



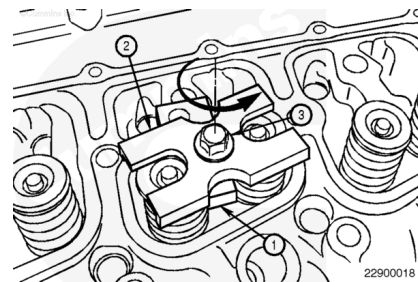
**⚠ CAUTION ⚠**

**Do not overtighten the capscrew. Doing so can damage the tool and/or the engine.**

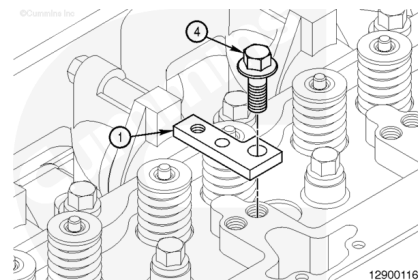
Turn the capscrew (3) **clockwise** until the collets can be removed. Use a magnetic tool.

Carefully turn the capscrew (3) **counterclockwise** until the tension is released from the valve springs.

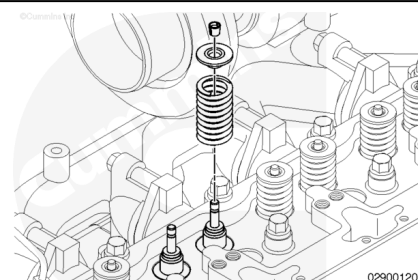
Remove the capscrew (3) and spring compression plate (2) from the lower support plate (1).



Remove the capscrew (4) and lower support plate (1).



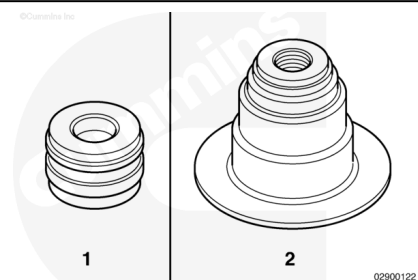
Remove the valve spring retainer and the valve spring.



**Note :** Prior to removing the valve stem seals, note the type of valve stem seal installed at each valve location. The same type of valve stem seal must be installed when assembling the cylinder head.

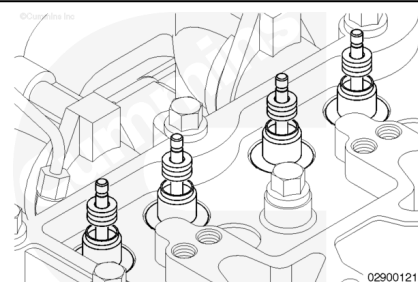
There are two types of valve stem seals used:

1. "Drive-On" Seal
2. "Top-Hat" Seal



Use boot pliers, Part Number 3163293, to remove the valve stem seal.

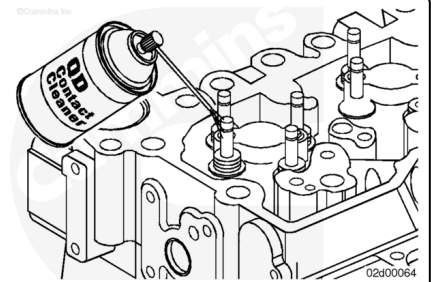
Remove and discard the valve stem seal.



## Clean and Inspect for Reuse

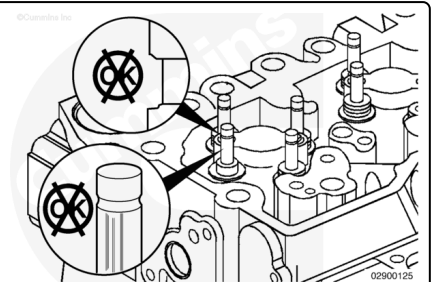
**Note :** The following Clean and Inspect for Reuse steps show illustrations of a 4 valve per cylinder head. The Clean and Inspect for Reuse steps for a 2 valve per cylinder are the same.

Clean the seal tower and valve stem with electrical cleaner, Part Number 3824510.



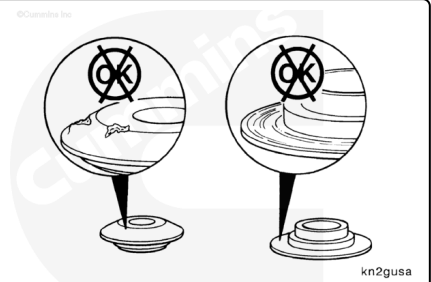
Inspect the exposed valve stem for scoring or heaving polishing. Inspect the valve collet grooves for wear.

If the valve stem is damaged, the cylinder head **must** be removed and the valve replaced. Refer to Procedure 002-004 (</qs3/pubsys2/xml/en/procedures/40/40-002-004-tr.html>).



Inspect the valve spring retainers and valve collets for damage or worn areas.

Discard and replace damaged and worn parts.



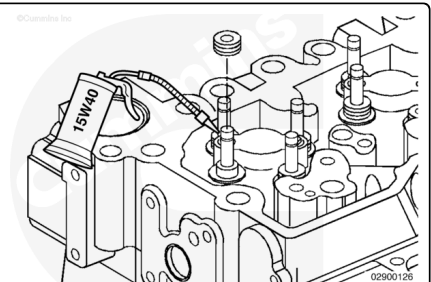
## Install

### ⚠ CAUTION ⚠

**Lubricate all the valve guide bores and valve stems with SAE 15W-40 engine oil. Failure to lubricate the valve guides and valve stems can result in premature valve guide wear.**

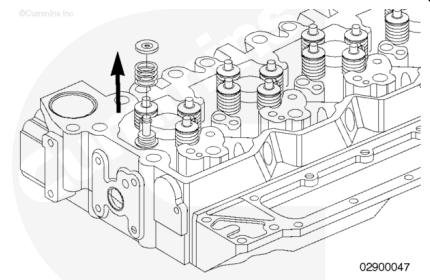
4 Valves Per Cylinder:

Lubricate the stems with SAE 15W-40 engine oil before installing the valve stem seals.



Install new valve stem seals. Use a valve stem installation tool, Part Number 3164055.

Install the valve spring retainer and valve springs.

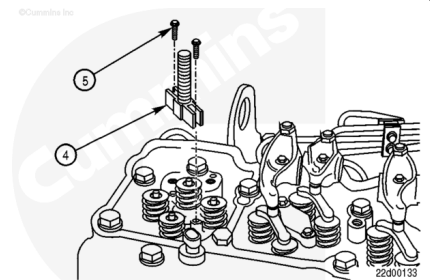


Compress the valve springs. Use the valve spring compressor, Part Number 3164329.

Position the replacer screw (4) above the injector bore and install the two capscrews (5) in the cylinder head where the hold-down clamp screws were removed.

Tighten the capscrews (5).

**Torque Value:** 5 n·m [ 44 in-lb ]



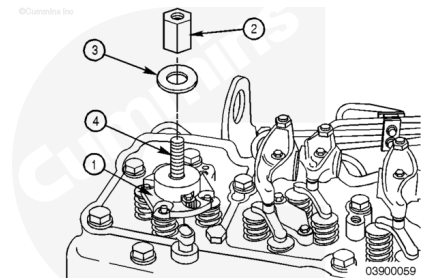
**Note :** The valves are not evenly spaced from the injector bore. It is important to align the slots in the valve spring compressor plate with the valve springs.

Apply anti-seize lubricant to the replacer screw (4) threads.

**Always** read and follow label precautions.

Position the valve spring compressor plate (1) on the replacer screw (4) and align the slots in the valve spring compressor plate with the valve springs.

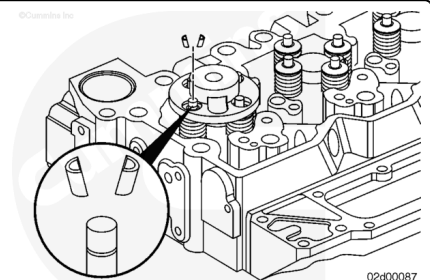
Install the washer (3) and nut (2) on the replacer screw (4).



### **⚠ WARNING ⚠**

**Valve springs are under tension and can act as projectiles if released. To reduce the possibility of eye injury, wear safety glasses with side shields.**

**Note :** Because there is a gap between the top of the piston and the valve face, it may be necessary to use a second magnet to pull the valve stem up to remove the valve collets.





Compress the valve springs until the valve collets can be installed.

Install the valve collets.

Service Tip: Use Assembly Lube, Part Number 3163087, or equivalent, on the valve collets to help hold them in place until the valve spring compressor is released.

Remove the valve spring compressor service tool.

**Note : Make sure the timing pin is pulled out before the engine is barred over to avoid damaging the timing pin.**

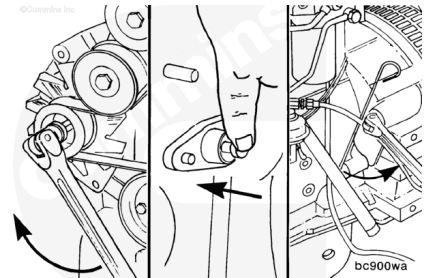
Use the marks made previously on the vibration damper/crankshaft pulley, rotate the engine **clockwise** to the next cylinder in the firing order.

Service Tip: If the barring tool is inaccessible, in some applications the alternator pulley can be used to bar the engine over.

Firing Order:

Six cylinder engines - 1-5-3-6-2-4

Four cylinder engine - 1-3-4-2

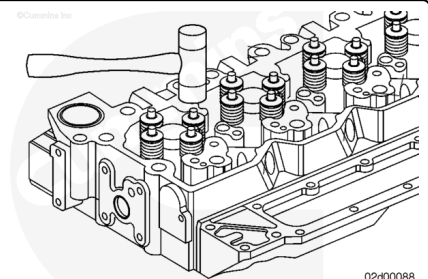


**⚠ WARNING ⚠**

**To reduce the possibility of personal injury, wear eye protection. If the collets are not correctly installed, they can fly out when the stems are hit with a hammer.**

**⚠ CAUTION ⚠**

**Rotate the engine to the next cylinder in the firing order before hitting the valve stem of the cylinder previously worked on. This will ensure that the valve does not contact the piston, resulting in a bent valve and internal engine damage.**

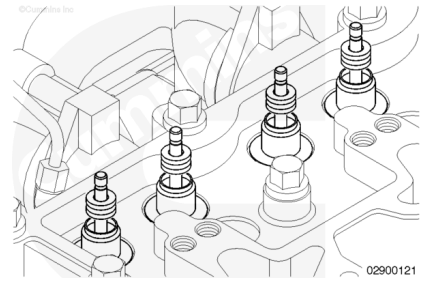


After rotating the engine to the next cylinder in the firing order, hit the valve stems of the cylinder previously worked on with a plastic hammer to make sure the collets are seated.

Repeat the previous steps until all of the valve stem seals are replaced.

**⚠ CAUTION ⚠**

**Failure to have the piston at top dead center will result in the valve dropping into the cylinder. If the valves drop into the cylinder, engine damage can result.**



**⚠ CAUTION ⚠**

**The same type of valve stem seal must be installed in the same location as removed. Incorrect valve stem seals will result in excessive oil consumption and internal engine damage.**

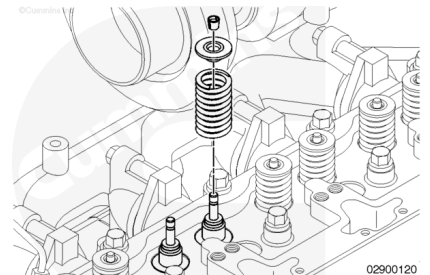
**⚠ CAUTION ⚠**

**Lubricate all the valve guide bores and valve stems with SAE 15W-40 engine oil. Failure to lubricate the valve guides and valve stems can result in premature valve guide wear.**

2 Valves Per Cylinder:

Lubricate the stems with SAE 15W-40 engine oil before installing the valve stem seals.

Install the valve springs and valve spring retainers.

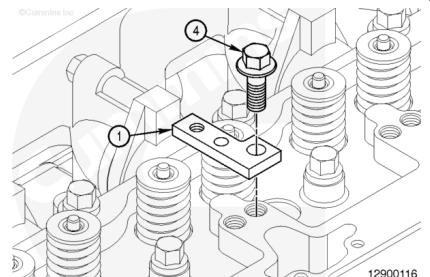


**⚠ CAUTION ⚠**

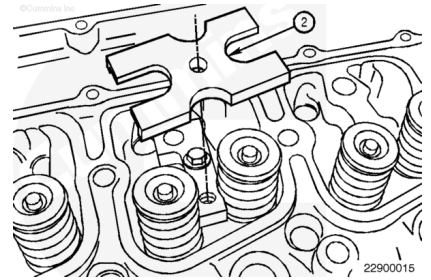
**Do not use an impact tool for any part of this procedure. Doing so can damage the tool and/or the engine.**

Compress the valve springs with valve spring compressor, Part Number 3163292.

Position the lower support plate (1) with the part number up, and install the 10-mm capscrew (4) through the plate and secure to the cylinder head. Tighten the capscrew finger-tight.

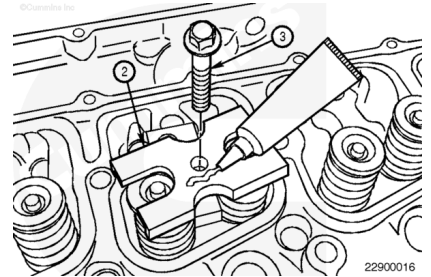


Position the spring compression plate (2) onto the valve springs with the part number up.



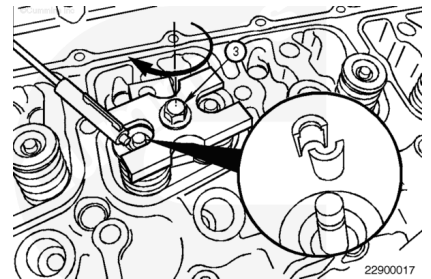
Apply shop grease to the top of the spring compression plate (2) and underneath of the capscrew (3) head and install the 10-mm (3) capscrew.

Tighten the capscrew finger-tight (3).



**⚠ WARNING ⚠**

**Wear appropriate eye and face protection when performing this procedure. Springs are under tension and can act as a projectile if released, causing personal injury.**



**⚠ CAUTION ⚠**

**Do not overtighten the capscrew. Doing so can damage the tool and/or the engine.**

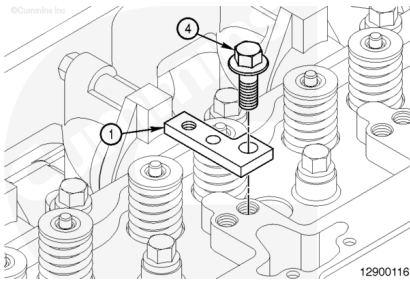
Turn the capscrew (3) **clockwise** until the collets can be installed. Use a magnetic tool.

Carefully turn the capscrew (3) **counterclockwise** until the tension is released from the valve springs.

Remove the capscrew (3) and spring compression plate (2) from the lower support plate (1).



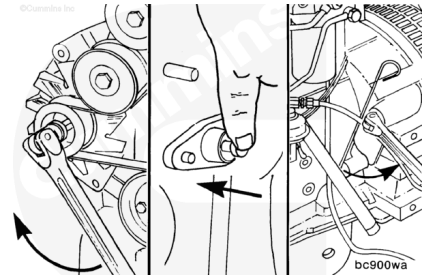
Remove the capscrew (4) and lower support plate (1).



**Note :** Make sure the timing pin is pulled out before the engine is barred over to avoid damaging the timing pin.

Use the marks made previously on the vibration damper/crankshaft pulley, rotate the engine **clockwise** to put the next group of cylinders at TDC.

Service Tip: If the barring tool is inaccessible, in some applications the alternator pulley can be used to bar the engine over.

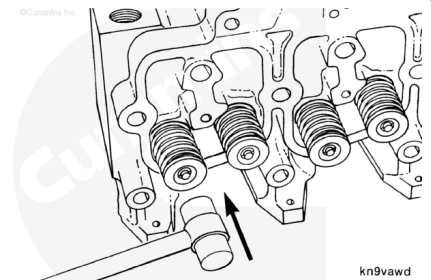


### **⚠ WARNING ⚠**

To reduce the possibility of personal injury, wear eye protection. If the collets are not correctly installed, they can fly out when the stems are hit with a hammer.

### **⚠ CAUTION ⚠**

Rotate the engine to the next cylinder in the firing order before hitting the valve stem of the cylinder previously worked on. This will ensure that the valve does not contact the piston, resulting in a bent valve and internal engine damage.



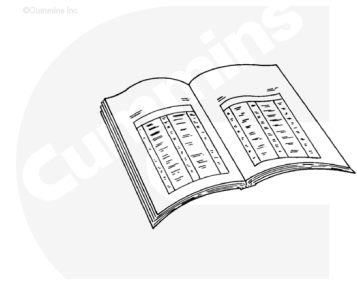
After rotating the engine to the next group of cylinders, hit the valve stems of the cylinders previously worked on with a plastic hammer to make sure the collets are seated.

Repeat the previous steps until all of the valve stem seals are replaced.

## Finishing Steps

**⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**



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- For engines with 2 valves per cylinder, install the rocker lever assemblies. Refer to Procedure 003-008 in Section 3 (</qs3/pubsys2/xml/en/procedures/40/40-003-008-tr.html>).
- For engines with 4 valves per cylinder, install the rocker lever assemblies and cross heads. Refer to Procedure 003-008 in Section 3 (</qs3/pubsys2/xml/en/procedures/40/40-003-008-tr.html>) and Refer to Procedure 002-001 in Section 2 (</qs3/pubsys2/xml/en/procedures/40/40-002-001-tr.html>).
- Adjust the overhead. Refer to Procedure 003-004 in Section 3. (</qs3/pubsys2/xml/en/procedures/40/40-003-004-tr.html>)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3 (</qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html>).
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine and check for leaks.